

Subhransu S. Bhattacharjee

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AI PhD candidate at the Australian National University researching computer vision, generative world models, and token-efficient large language model and vision-language model architectures for multimodal and spatial reasoning. First-author CVPR 2025 and ECCV 2026 researcher with applied-science experience at Microsoft, Optiver, and JPMorgan.

Education

Doctor of Philosophy

School of Computing, Australian National University, Australia

Artificial Intelligence

Apr 2023 – Mar 2027 (expected)

- **Thesis topic:** *Into the Unknown: Tackling Spatio-semantic Uncertainty using Generative Posteriors*

- **Supervisors:** Dr. Rahul Shome, Dr. Dylan Campbell, and Prof. Stephen Gould

Bachelor of Engineering (Honours)

School of Engineering, Australian National University, Australia

Mechatronics

Jul 2018 – Dec 2022

- **First Class Honours;** GPA 6.48/7.00; ranked 3rd in cohort. **Minors:** Mathematics; Electronic Communication Systems.

- **Machine Learning,** London School of Economics Summer School — **Grade A.**

- **Thesis project:** [Whiplash Gradient Descent Dynamics](#).

Industry Research Experience

Microsoft Corporation, India

May 2026 – Jul 2026

Applied Scientist PhD Research Intern, LLM-as-Judge, RAG & Agent Evaluation

- Built an end-to-end LLM-as-judge framework for RAG and agent outputs in collaboration with Microsoft India Development Center, Microsoft Research India, and Microsoft AI (MAI), combining groundedness, relevance-purity, response-discipline, and dissatisfaction-aware evaluation.
- Developed an agentic ingestion harness that converts unstructured, Likert-scored annotations into structured evaluation records, reducing annotator cognitive load by $\sim 35\%$ and increasing usable annotation throughput $\sim 2\times$, supporting golden-set construction, evaluator calibration, and failure analysis.

Optiver, Sydney, Australia

Nov 2024 – Feb 2025

Quantitative Research Intern, Neural-Fuzzy Modelling, xLSTM, XGBoost, SHAP

- Developed a neural-fuzzy joint classifier for Korean retail-flow prediction, achieving **94% classification accuracy**, integrating xLSTM and XGBoost within a streaming market-feature pipeline with online SHAP attribution.
- Built leakage-aware walk-forward evaluation and model-monitoring infrastructure for Korean market microstructure, and separately developed and backtested a statistical-arbitrage model for the Hong Kong market.

JPMorgan Chase & Co., Sydney, Australia

Dec 2021 – Feb 2022

Quantitative Research Intern, NLP, Information Retrieval, Risk

- Developed a metadata-indexing and retrieval pipeline over 10,000+ financial documents supporting transformer-based question answering and structured answer extraction for risk-research workflows, improving top- k retrieval recall by ~ 8 percentage points.

Selected Publications

Subhransu S. Bhattacharjee, Dylan Campbell & Rahul Shome: *Believing is Seeing: Unobserved Object Detection using Generative Models.* *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, Nashville, Tennessee, USA. [Project page](#). Defines 2D, 2.5D, and 3D unobserved object detection from a single RGB view with diffusion/VLM pipelines and tailored metrics; the 3D-diffusion method outperforms all applicable methods, including an in-frustum oracle, on the 3D task.

Subhransu S. Bhattacharjee, Dylan Campbell & Rahul Shome: *FlatLands: Generative Floorplan Completion From a Single Egocentric View.* *European Conference on Computer Vision (ECCV)*, 2026, Malmö, Sweden. Springer. [Paper](#).

Releases FlatLands: 270,575 observations from 17,656 real metric indoor scenes across six datasets.

Compares 11 methods, showing stochastic generators improve completion fidelity and capture layout uncertainty.

Subhransu S. Bhattacharjee, Hao Lu, Dylan Campbell & Rahul Shome: *MatterDoor: Sampling Zero-shot Spatio-semantic Priors using Generative Models.* *arXiv:2510.11014*, 2025. Under review; RSS 2026 FM4RoboPlan workshop presentation, Sydney. [Preprint](#) | [RSS Workshop Paper](#).

Samples environment priors from partial observations using pretrained generative models.

Produces RGB-D point-cloud hypotheses for occupancy, target semantics, and configuration-space planning.

Subhransu S. Bhattacharjee & Ian Petersen: *Analysis of the Whiplash Gradient Descent Dynamics.* *Asian Journal of Control* (Special Issue), Wiley, 2023. DOI: [10.1002/asjc.3153](https://doi.org/10.1002/asjc.3153).

Introduces a closed-loop inertial optimiser with adaptive damping, deterministic saddle-point escape, and symplectic Lyapunov convergence analysis. Establishes polynomial and exponential convergence rates for adaptive inertial gradient dynamics. [Code](#).

Conference papers: ASCC 2022, Jeju Island (DOI); ANZCC 2021, Gold Coast (DOI).

Open-Source Research & Software

TorchKAN — Independent PyTorch implementation of Kolmogorov–Arnold Networks with spline, Legendre (KAL-Net), Chebyshev (KAC-Net), and convolutional (KANvolver) variants; classification and regression experiments, CUDA execution, post-training quantisation, and Integrated-Gradients interpretability. Sole maintainer. [Project Site](#) | [Code](#)

Mangroves — Python package for depth-based hierarchical data management: type-aware ingestion, dynamic attribute access, result caching, and CPU–GPU tensor movement for episodic training pipelines. Sole author; maintenance-only. [Project Site](#) | [Code](#)

TuringTree — Vectorless, reasoning-based RAG running fully on-device (Ollama + Qwen) with 0–100 confidence scoring and abstention; tests, CI, and a tagged release. Co-developed by a four-person team during the Microsoft Global Intern Hackathon. [Project Site](#) | [Code](#)

Research Experience

Research School of Management, Australian National University

Sep 2023 – Sep 2024

Graduate Research Assistant — FinTech & AI, PI: Dr Priya Muthukannan

- Developed AI-adoption assessment frameworks for an industry-linked Commonwealth Bank of Australia project, translating open-banking and dynamic-capabilities analysis into strategy recommendations.

CIICADA Lab, Australian National University

Dec 2022 – Feb 2023

Summer Research Scholar — Control & Optimisation, Supervisor: Prof Ian Petersen

- Advanced feedback-based optimisation analysis for systems without closed-form solutions, contributing to publications at ASCC 2022 and ANZCC 2021.

School of Computing, Australian National University

Mar 2022 – Jun 2022

Undergraduate Research Scholar — Geometry of Data, Supervisor: Prof Richard Hartley

- Tested invertibility of differentiable neural mappings with dense positional encoding, achieving a 72% hit rate under an RMSE criterion.

Armament Research & Development Establishment (ARDE), Pune, India

May – Jul 2020

Summer Research Trainee — Passive Radar Signal Processing, Kalman Filtering, FPGA, Real-Time DSP

- At ARDE, a laboratory of India's Defence Research and Development Organisation (DRDO), developed a Kalman-filter-centred method for selecting matched filters for noisy passive-radar signals and an FPGA-based multiprocessor interface for real-time long-range radar analysis.

Awards & Recognition

2026 DARPA Travel Grant (AUD 6,000) for the European Conference on Computer Vision (ECCV 2026).

2026 Research contributor on a Google Research grant supporting ANU Robotics work on world models for task-and-motion planning.

2025 Vice-Chancellor's Travel Grant, Australian National University, for CVPR 2025 travel.

2024 Optiver PhD Quant Lab Program (selected participant).

2023 ANU International University Research Scholarship (4 years) and Higher Degree by Research Merit Stipend.

2022 Highly Recommended Paper at the *Asian Control Conference*; Best Student Reviewer Award.

2021 High Commendation Award for undergraduate student paper at the *Australia & New Zealand Control Conference*.

2020 ANU Chancellor's International Scholarship — undergraduate tuition-fee award for academic merit.

Selected Invitations & Talks

2026 Invited attendee at the Gemma release hosted by Google DeepMind, Hyderabad.

2026 Selected invitee to the Hugging Face, PyTorch, and Red Hat Meetup, Bengaluru.

2025 Invited research talk at the Dyson Robotics Laboratory, Imperial College London.

2025 Invited research presentation at the Citadel & Citadel Securities PhD Summit, London.

Teaching & Service

Senior/Head Tutor, Australian National University: Programming Fundamentals; Introduction to Machine Learning; Building Cyber-Physical Systems; Responsible Leadership; Power Systems and Devices; Network Control & Optimization; Advanced Control Theory.

Reviewing (2021–2026): CVPR, ECCV, ACCV, DICTA, AJCAI, NeurIPS, ICML, AAAI, ICRA, IROS, ACC, IEEE T-RO/RA-L, Asian Journal of Control.

Technical Skills

Programming & Scientific Computing: Python; PyTorch (Autograd, `torch.compile`/AOTAutograd); Triton; SQL; C; MATLAB; Bash; NumPy; SciPy; pandas; scikit-learn.

Computer Vision, Generative & Spatial AI: generative world models; diffusion; flow matching; 3D scene understanding; uncertainty estimation; OpenCV; Open3D; PyTorch3D; COLMAP; ROS2.

LLM/VLM Architecture & Inference: token pruning/merging/compression; FlashAttention; KV-cache optimisation; sparse/MoE routing; quantisation; speculative decoding; vLLM; hardware-aware co-design; LLM-as-judge; RAG/agent evaluation.

Quantitative Methods: statistical inference; time-series modelling; feature engineering; market microstructure; signal research; regime-aware analysis; walk-forward validation; backtesting; risk modelling; XGBoost; xLSTM; SHAP.

ML Systems & Infrastructure: mixed-precision/multi-GPU training; inference profiling; benchmarking; reproducible pipelines; Linux; Git; Docker; SLURM; CUDA; AWS (EC2, S3); Weights & Biases.